



Temperature gradient and shear strength in nanosecond laser welding of silicon carbide and copper

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ARTICLE INFO

Handling Editor: P Rios

Keywords:

Nanosecond laser welding
Silicon carbide (SiC)
Heat-affected zone (HAZ)
Temperature gradient

ABSTRACT

This study investigates the bonding and cracking behavior of SiC–Cu joints during nanosecond laser welding, focusing on the influence of temperature gradients and processing parameters. By combining experiments and simulations, optimal bonding conditions were identified: laser power between 35 and 45 W and scanning speed of 2–4 mm/s, producing temperature gradients of 22,000–35,000 K/cm. Within this range, crack-free bonding occurred with a width-to-depth ratio ~ 0.967 , while higher ratios (~ 1.014) led to excessive lateral heat spreading, stronger thermal stresses, and consequently higher crack susceptibility. The heat-affected zone (HAZ) was minimized to 10–25 μm , and EDS analysis revealed a Cu–O–Si interfacial layer, mitigating the formation of brittle intermetallic compounds (IMCs). The highest shear bonding strength of 4.5 MPa was achieved at the laser power of 45 W and scanning speed of 3 mm/s, emphasizing the significance of precise parameter optimization. In contrast, at laser powers exceeding 46 W, temperature gradients surpassed 37,000 K/cm, produced severe thermal stresses and interfacial instability, HAZ expansion, and crack formation. These conditions reduced the shear strength to as low as 1 MPa. The research provides critical insights into temperature-gradient management and bonding reliability, supporting the development of robust semiconductor–metal joints for microelectronics, aerospace, and high-performance energy systems.

1. Introduction

Laser welding has become a transformative technology in modern manufacturing, offering high precision and efficiency for material joining across various industries. Its ability to modify materials at microscopic scales with minimal impact on surrounding areas significantly expands its applications in high-precision manufacturing [1]. However, welding dissimilar materials, such as aluminum–copper (Al–Cu) and aluminum–steel (Al–Steel), presents unique challenges due to differences in melting points, thermal conductivities, and the formation of brittle IMCs, which can degrade joint strength and performance [2–6]. In Al–Cu laser welding, pulsed Nd:YAG laser welding techniques optimize pulse shape and pulse overlap ratio to achieve high tensile strength and low electrical resistance. Advanced methods, including spatial beam oscillation and acoustic emission monitoring, further enhance weld quality by ensuring precise seam control and enabling real-time assessment [4,5,7]. Similarly, in Al–Steel welding, beam offset toward the aluminum side is an effective strategy to reduce IMC thickness, while the use of interlayers mitigates brittle phase

formation, improving overall joint strength [6]. Additionally, optimizing key parameters such as beam positioning, welding speed, and power remain crucial for maintaining mechanical integrity and controlling IMC formation in dissimilar material welds [4–6]. Xia et al. [8] conducted an in-situ SEM tensile investigation on Al/Steel interfacial fracture behavior, revealing that a controlled 2–3 μm $\tau 5\text{-Fe1.8Al7.2Si}$ IMC layer, formed under optimized thermal conditions (968°C–1003 °C with 12 wt% Si content), provided an optimal balance between joint strength and crack resistance.

Recent studies have demonstrated the feasibility of laser welding for joining metal and glass. To enhance weld quality and minimize residual stress, techniques such as oxide coating and post-process annealing are commonly employed [9–11]. These methods help maintain the integrity and durability of welded components, thereby improving the effectiveness of laser welding for dissimilar materials. However, the significant difference in thermal expansion rates between metal and glass remains a major challenge [12,13]. To address these challenges, researchers have explored ultrafast laser technology as a promising solution. Carter et al. [14] demonstrated the effectiveness of ultrafast laser welding for

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<https://doi.org/10.1016/j.jmrt.2025.10.034>

Received 11 May 2025; Received in revised form 2 October 2025; Accepted 3 October 2025

Available online 5 October 2025

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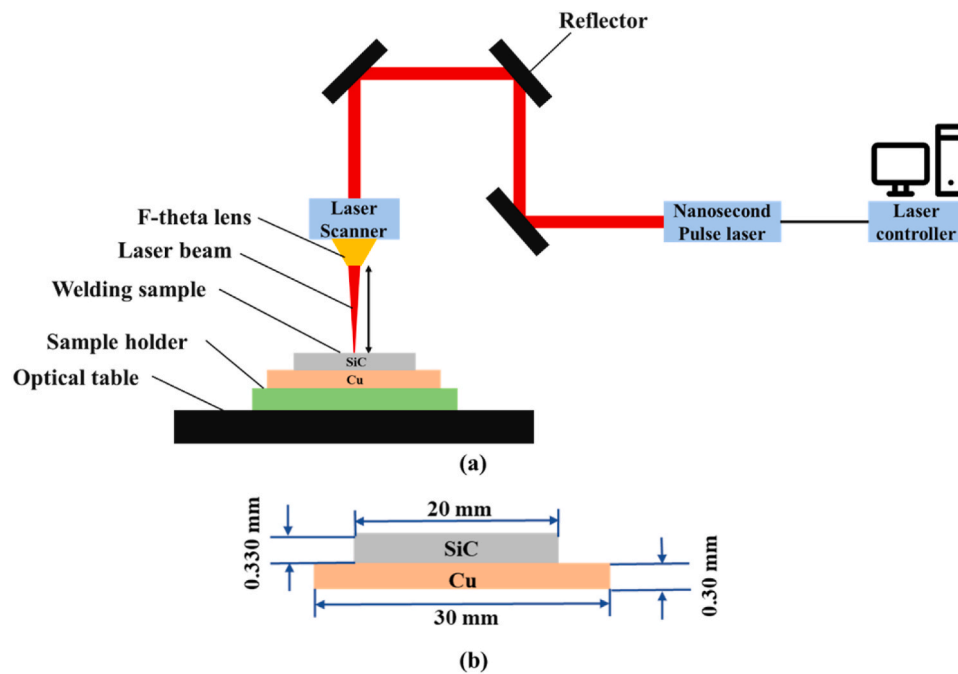


Fig. 1. (a) Schematics of the laser welding setup, and (b) Schematic illustration of SiC–Cu geometry.

dissimilar materials, including metal and glass. Wang et al. [15] further elucidated the mechanisms underlying the welding of stainless steel to glass, highlighting key processes such as wetting, adsorption, and mechanical bonding. Additionally, Ozeki et al. [16] successfully joined copper with non-alkali glass using an infrared femtosecond laser, demonstrating a precise and efficient approach to material bonding. The role of laser-induced metal particles in facilitating strong bonds has also been investigated. In addition, Zhang et al. [17] emphasized the critical role of metal particles generated through ultrafast laser ablation in welding alumina-silicate glass to copper. Furthermore, Li et al. [18] showed that effective welding of glass to copper requires melting the glass to fill gaps, ensuring strong adhesion and joint integrity. This melting process is essential for achieving durable welds and enhancing the reliability of laser-welded glass-metal joints. Recent breakthroughs in this field involve adapting ultrafast laser techniques into the infrared spectrum, enabling precise silicon-metal welding with shear strengths up to 2.2 MPa, crucial for microelectronics applications [19]. Also, Sopena et al. [2] demonstrated that nanosecond laser pulses achieved robust shear strengths of 32 ± 10 MPa for dissimilar semiconductor materials, such as gallium arsenide and silicon. For transparent materials, nanosecond laser welding effectively addresses the challenge of achieving robust and reliable bonding without intermediate light-absorbing layers or adhesives. The precision and control provided by nanosecond pulses enable effective bonding while minimizing thermal effects, making this technique ideal for high-precision manufacturing applications involving transparent and delicate materials [13,20].

It is noted that temperature gradients, material microstructure, and cooling rates influence crack formation during laser beam welding (LBW). The high energy density and rapid heating in LBW create steep temperature gradients, resulting in significant thermal stresses in the weld zone. Combined with fast cooling rates, these stresses can promote hard, brittle phases like martensite in certain alloys [21]. Adjusting welding parameters, such as laser power and scanning speed, affects heat input, influencing weld seam characteristics and temperature distribution [22,23]. However, precise measurement of temperature variations remains challenging due to the rapid heating and cooling involved [24]. To address this issue, numerical simulations have been developed to model heat distribution and thermal cycles during LBW.

These simulations provide valuable insights into optimal parameter settings that minimize crack formation. They utilize advanced heat source models to accurately simulate temperature gradients, predicting conditions under which cold cracking may occur, such as steep thermal gradients, rapid solidification, and localized tensile stresses that exceed the joint's fracture resistance [25,26].

Recent studies further indicate that micro-cracking is not governed by thermal gradients alone; chemistry- and microstructure-driven changes at grain boundaries—such as elemental segregation, complexion transitions, and precipitation state—can strongly modulate deformation pathways and crack susceptibility under rapid thermal cycling [27–29]. For example, thermal-cycle-induced precipitate dissolution and reduced stacking-fault energy have been shown to alter twinning/slip behavior and plasticity, while multiscale analyses reveal that nanoprecipitation and dislocation cell structures can intensify strain localization [27]. Liu et al. [29] studied NiTi alloys that have also linked dislocation slip and martensite stability to irreversible deformation. These insights suggest that a comprehensive understanding of joint reliability requires not only thermal gradient control, but also microstructural interrogation of segregation and precipitation behavior.

To address this, the present work integrates experimental analysis and FLOW-3D simulations to investigate nanosecond laser welding of SiC to Cu, with a particular focus on temperature-gradient-driven cracking, melt pool geometry, and interfacial behavior. An artificial neural network (ANN) is also employed to map optimal processing windows. The objectives of this study are to elucidate the thermo-mechanical mechanisms governing SiC–Cu bonding, determine a critical width-to-depth ratio for achieving crack-free welds, quantify the shear strength under varying processing parameters, and assess interfacial microstructural evolution. These findings aim to support the development of reliable bonding strategies for semiconductor packaging and related applications. This work is distinct from our earlier study [30], which focused on process optimization of Cu–SiC lap joints using ANN modeling and wobbling strategies. In contrast, the present study investigates a SiC-on-top configuration, where decomposition behavior and thermal transport effects play a decisive role in crack formation. Together, these two studies are complementary: the previous work provides optimization pathways, while the current one advances the understanding of thermo-mechanical reliability in SiC–Cu joints.

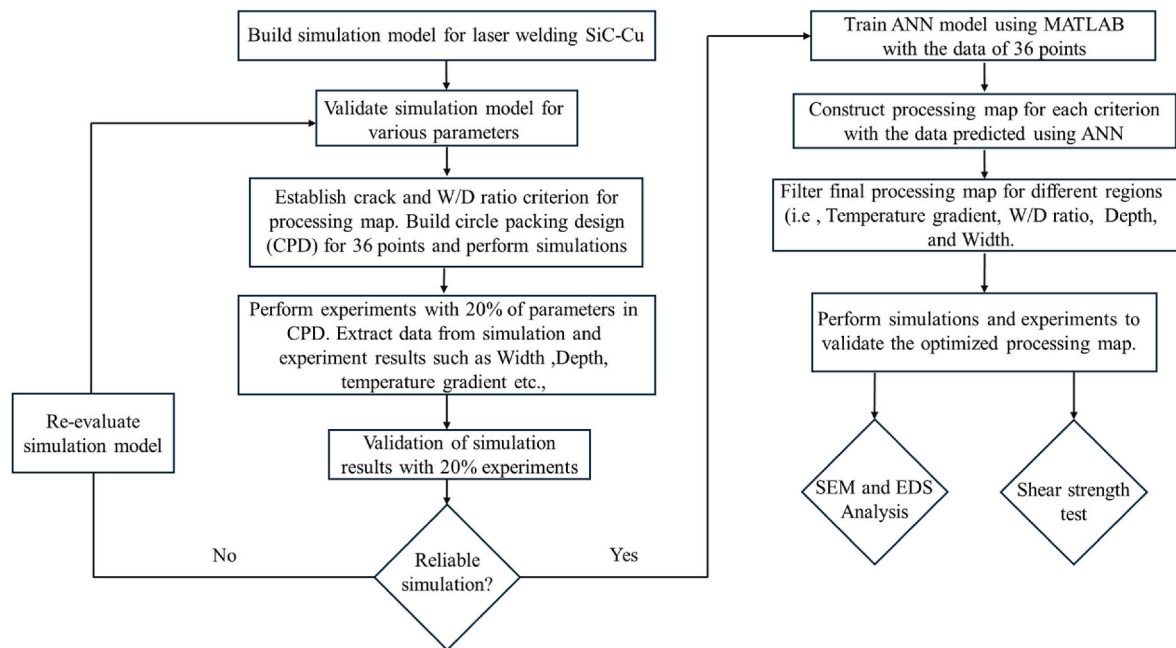


Fig. 2. Flowchart illustrating the research methodology, including simulation setup, experimental trials, and final validation at eight representative parameter points selected from the optimized processing map.

2. Materials and methods

In the experiment, a 4H-SiC substrate (0.33 mm thick) was joined to a commercially pure copper foil (0.30 mm thick) using an overlapping laser welding technique. As depicted in Fig. 1(a), the welding process utilized a nanosecond ytterbium fiber laser (model: IPG Photonics YLPN-1000-4x200-30-M), operating at a wavelength of 1064 nm. The laser parameters, such as pulse repetition frequency (45–65 kHz) and pulse duration (120 ns), were carefully optimized to deliver an energy density adequate for achieving a robust bond between the materials. In addition, the focal length (125 mm), laser spot diameter (15 μm), and beam quality ($M^2 < 1.5$) were precisely adjusted to focus the laser energy accurately on the substrate's surface. These parameters were established through comprehensive experimental trials, ensuring a consistent and high-quality weld, underpinning the experimental outcomes' reliability and credibility. Furthermore, deliberate variations in the processing parameters, specifically the scanning speed and average laser power, were systematically introduced to evaluate their influence on the welding process and outcomes.

It should be noted that no direct experimental measurement of temperature was carried out in this study. Instead, thermal behavior was analyzed through Flow-3D simulations, which provided temperature gradients under different welding conditions. These temperature gradients were used to evaluate thermal behavior and to construct processing maps.

The SiC-Cu lap-joint configuration, as depicted in Fig. 1(b), was utilized for practical trials with a modified arrangement where the SiC substrate is positioned on top of the copper sheet. This alteration was informed by thoroughly understanding these materials' optical and thermal properties. Specifically, SiC's transmittance at the 1064 nm wavelength used in laser welding is only 18.2 % [31], meaning that although most of the laser energy is absorbed by SiC, a portion of it can still be transmitted. In this configuration, our approach aims to partially melt the SiC layer to achieve a stable joint while mitigating the risk of microcracks typically associated with direct welding on SiC. Arranging SiC as the top layer in direct contact with the copper sheet provides an effective thermal regulation system, leveraging copper's role as a heat sink [32]. This configuration, combined with Flow-3D predictions, confirmed that controlled thermal transfer reduced the temperature

gradient across the SiC layer, minimizing the risk of thermal shock and enabling stable interfacial fusion. This staged energy transfer promotes a gradual and consistent temperature increase within the SiC substrate, thereby minimizing sharp temperature differences and reducing the possibility of interfacial defects or inconsistencies. The process facilitates partial melting of the SiC substrate through precisely calibrated heat input, promoting stable interfacial fusion and minimizing thermal stress. Thus, this approach enables the formation of defect-free metal–ceramic joints with improved mechanical integrity and reduced crack susceptibility.

The key steps involved in this study are illustrated in the flowchart (Fig. 2), providing a clear and structured understanding of the methodology. The diagram outlines the sequential process, including data collection, simulation, ANN modeling, and experimental validation. The description of processing map construction has been expanded: the maps were generated using Flow-3D temperature-gradient simulations and ANN predictions and then filtered by excluding parameter regions associated with excessive heat input. The acceptable bonding region was therefore defined by combining both simulation outputs and mechanical testing. Sample characterization was performed using both optical microscopy and scanning electron microscopy (SEM). Optical microscopy was applied to examine weld and HAZ dimensions, while SEM coupled with energy-dispersive spectroscopy (EDS) was employed to investigate fracture surfaces and interfacial microstructures. These techniques provided direct experimental validation of joint quality and microstructural features.

2.1. Interface-specific placement of temperature gradient probes

In this study, temperature gradients were analyzed at various average laser powers and scanning speeds to assess their effects on cracking and bonding in SiC-Cu joints. Our model accounts for material diffusion and laser absorption, requiring precise temperature monitoring. During simulations, temperature probes were strategically placed at the SiC-Cu interface to ensure accurate evaluation of thermal gradients without affecting simulation results. Placing temperature probes at the interface rather than on the SiC surface is critical because this region is where heat transfer, material diffusion, and bonding occur. The temperature gradient probes at the interface directly influence the

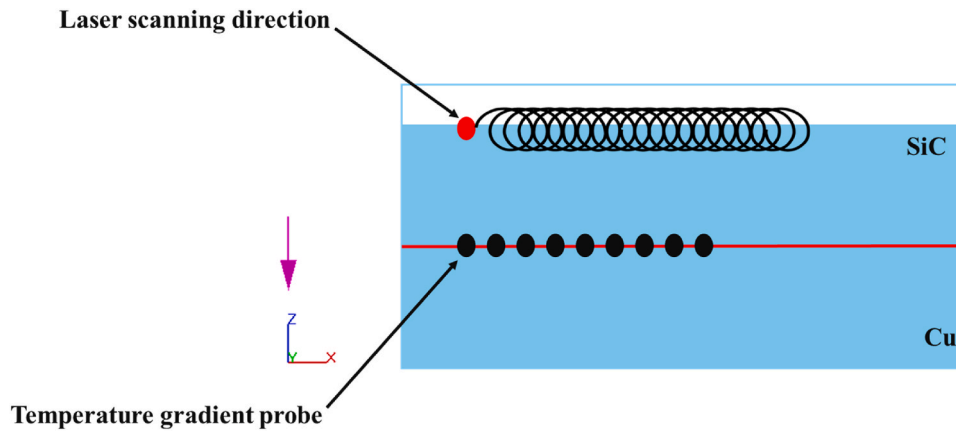


Fig. 3. Schematic of the laser scanning direction and placement of eight virtual temperature probes along the SiC–Cu interface in the FLOW-3D simulation. Multiple probes were used to capture non-uniform temperature distributions caused by wobbling scanning.

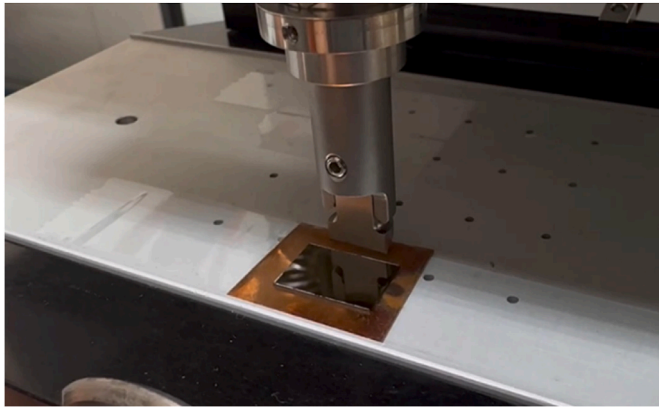


Fig. 4. Shear test setup using the STELLAR 4000 Die Shear Tester. The cartridge shear mechanism applies a controlled load to the bonded SiC–Cu specimen to evaluate interfacial shear strength.

thermal expansion mismatch between SiC and Cu; excessive gradients can cause thermal stress concentration and crack formation, compromising joint integrity. Moreover, we employed FLOW-3D to simulate laser welding, focusing on the effects of average laser power and scanning speed on melting pool dimensions and temperature gradient. The simulation captures transient heat transfer and fluid flow to analyze weld formation and stability.

The wobbling laser strategy aims to achieve uniform heat distribution, and interface probes help verify whether heat is effectively dispersed to prevent localized overheating and ensure strong metallurgical bonding. As a brittle ceramic, SiC is highly sensitive to thermal stresses induced by sharp temperature gradients, which can lead to cracks and compromise joint integrity. Since the SiC–Cu interface is crucial for bonding strength, analyzing this region in simulations helps assess cracking behavior and determine optimal metallurgical bonding conditions while maintaining structural integrity.

As shown in Fig. 3, the simulation was designed to probe temperature gradients and evaluate the thermal effects of the dynamic wobbling laser strategy on the SiC–Cu interface. Fig. 3 presents the SiC–Cu interface with eight strategically placed temperature probes (blue dots). These probes are arranged perpendicular to the interface to monitor temperature distribution and assess thermal gradients. Temperature gradients were extracted from the simulation domain using probes positioned exactly at the SiC–Cu interface, one on the SiC side and one on the Cu side, aligned vertically through the center of the molten pool. In Fig. 3, the probe spacing was set at 10 μm intervals along the x-axis,

allowing calculation of the local temperature gradient ($\partial T/\partial z$) across the SiC–Cu interface. This configuration ensured that the measured gradients captured the steepest thermal transitions between the brittle ceramic and ductile metal regions during cooling. A single temperature probe cannot accurately capture the dynamic thermal effects of wobbling scanning, so multiple probes provide comprehensive temperature distribution data, ensuring precise and reliable analysis. This probe configuration facilitates a detailed understanding of interfacial heat transfer and offers key insights into bonding mechanisms, joint integrity, and cracking behavior under different laser processing conditions.

2.2. Die shear test for bonding strength

This study assessed the shear strength of SiC–Cu lap joints bonded via laser welding using the STELLAR 4000 Die Shear Tester. The instrument's cartridge shear mechanism, capable of exerting up to 10 kg of load, was employed to evaluate the bonded samples (shear test setup shown in Fig. 4). Each SiC–Cu specimen was carefully positioned on the test stage to ensure accurate alignment and controlled force application, thereby minimizing experimental error.

Ultimately, eight representative samples, combining different laser power and scanning speed settings, were tested, with three measurements performed for each condition. This design enabled systematic evaluation of bonding strength across a range of energy inputs, allowing both optimal and crack-prone regions to be identified. The use of repeated measurements enhanced statistical reliability. This methodology provided a comprehensive assessment of mechanical stability and repeatability, offering direct insights into the influence of laser processing parameters on bond strength.

3. Numerical model set up

The formulation of the computational fluid dynamics (CFD) model relies on specific assumptions and governing equations. One primary assumption is that the laser beam's energy distribution follows a Gaussian pattern characterized by an axially symmetric Gaussian profile. Consequently, the heat flux q for laser welding within this CFD framework is expressed [33] as:

$$q = Q \cdot \exp \left[- \left(\frac{r}{r_b} \right)^2 \right] \quad (1)$$

where Q represents the peak heat density at the laser beam center, r is the radial distance, and r_b is the laser spot radius at the focal point. This Gaussian heat flux distribution is fundamental to the laser welding model, establishing a spatial heat input that creates temperature

gradients radiating outward from the laser beam's center and defining the material's temperature profile.

Ray tracing technology is applied to simulate laser paths, incorporating a real-time multi-reflection mechanism embedded within the Flow3D® system. Concurrently, the Volume of Fluid (VOF) method monitors the evolution of the weld pool's free surface dynamics during the process. The mass conservation equation is subsequently defined as [34]:

$$\nabla \cdot \mathbf{v} = \frac{\dot{m}_s}{\rho} \quad (2)$$

Here, the velocity vector $\mathbf{v}$ is influenced by the mass source term $\dot{m}_s$ originating from the metal droplet. This relationship shows that fluid velocity adjusts with the mass input rate, ensuring the proper integration of added material into the weld pool. This adjustment is critical for accurately simulating material movement and distribution in the weld pool, which is essential for tracking how metal flows and behaves during welding [34]. The energy conservation equation is presented as follows:

$$\frac{\partial h}{\partial t} + \mathbf{v} \cdot \nabla h = \frac{1}{\rho} \nabla \cdot (k \nabla T) \quad (3)$$

The relationship between enthalpy h and temperature T describes energy transfer dynamics, incorporating thermal conductivity k and material density ρ . The term $\nabla \cdot (k \nabla T)$ heat conduction driven by temperature gradients, is central to calculating spatial and temporal temperature distributions. This energy conservation equation forms the foundation for understanding heat flow and dissipation within the weld pool. It emphasizes how temperature gradients influence the enthalpy distribution, capturing the complex interactions that govern energy flow in the system. The model captures the phase transition behavior through an entropy-driven continuum framework, as shown in Eq. (4) [34]:

$$h = \begin{cases} \rho_s C_s T & (T \leq T_s) \\ h(T_s) + h_{sl} \frac{T - T_s}{T_l - T_s} & (T_s \leq T \leq T_l) \\ h(T_l) + \rho_l C_l (T - T_l) & (T_l \leq T) \end{cases} \quad (4)$$

In this formulation, ρ_s and ρ_l represent the solid and liquid phase densities, respectively. The parameters C_s and C_l denote the specific heat capacities per unit volume for the solid and liquid phases, while T_s and T_l mark the solidus and liquidus temperatures, indicating the onset and completion of melting. The term h_{sl} represents the latent heat associated with the melting/solidification of Cu and the high-temperature mass loss of SiC (laser-induced decomposition/evaporation). No additional solid-solid transformations are invoked; the corresponding energy exchange is captured through the enthalpy source terms and the evaporation/recoil-pressure model. This formulation relates enthalpy changes to temperature variations in the solid and liquid states, incorporating temperature gradients during phase transitions in the weld pool. Enthalpy adjustments reflect heat absorption or release due to phase shifts, influencing the material's thermal profile. The fluid volume equation, essential for tracking the motion of the keyhole's free surface, is expressed as follows [34]:

$$\frac{\partial F}{\partial t} + \nabla \cdot (\mathbf{v} F) = \dot{F}_s \quad (5)$$

Equation (5) describes the fluid volume dynamics within the weld pool, where F represents the fluid volume fraction, and $\dot{F}_s$ denotes the rate of change linked to the mass source term $\dot{m}_s$. The rate of change refers to the time derivative of the mass fraction undergoing phase transition, which is directly linked to the mass source term in the enthalpy-based energy equation. This formulation is essential for modeling the weld pool surface evolution, offering insights into fluid flow and movement within the molten zone and enabling precise tracking of keyhole dynamics.

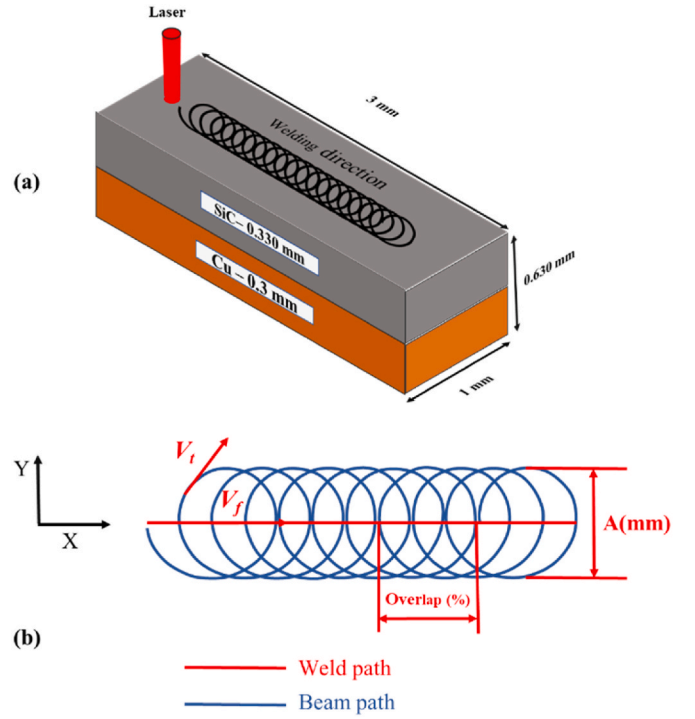


Fig. 5. (a) Schematic of Computational Model Domain for Simulation, (b) Schematic of the laser wobbling strategy showing beam oscillation pattern, tangential velocity, and linear travel speed, which together determine the resultant heat input profile.

The model takes into account the recoil pressure (P_R) that arises from the evaporation of molten metal at the keyhole walls, which significantly influences the keyhole dynamics, as represented in Eq. (6) [34]:

$$P_R(T) \cong 0.54 P_{sat}(T) \cong 0.54 P_0 \exp \frac{T - T_b}{RTT_b} \quad (6)$$

In this formulation, P_{sat} denotes the saturation vapor pressure, P_0 refers to the standard atmospheric pressure, L_v is the latent heat of vaporization, T_b is the material's boiling point, and R is the universal gas constant. This equation establishes the relationship between temperature and recoil pressure, where increases in temperature near the keyhole wall contribute to recoil pressure that impacts the molten material flow. This temperature-dependent mechanism influences the keyhole's stability and shape during welding.

The boundary conditions implemented in this simulation model are detailed as follows. The top and bottom boundaries of the computational domain are modeled with outflow conditions that account for a range of thermal processes, including heat conduction, convection, radiation, and evaporation. These processes are captured in Eq. (7). On the other hand, the side boundaries are specified with continuity conditions, as described in Eq. (8) [35].

$$\left\{ \begin{aligned} k \frac{\partial T}{\partial \vec{n}} &= -h_c(T - T_0) - \epsilon_r \sigma (T^4 - T_0^4) - \rho V_{evap} T \\ k \frac{\partial T}{\partial \vec{n}} &= -h_c(T - T_0) - \epsilon_r \sigma (T^4 - T_0^4) \end{aligned} \right. \quad (7)$$

Equation (7) defines the thermal boundary conditions as follows: Top surface subjected to a laser heat flux applied along the surface normal vector $\vec{n}$, combined with convective heat losses to the surrounding air and radiative heat losses to the environment. Bottom surface — in thermal contact with the copper substrate held at 300 K, allowing heat conduction into the substrate and supplemented by convective cooling to the ambient air. This boundary setup is critical for maintaining

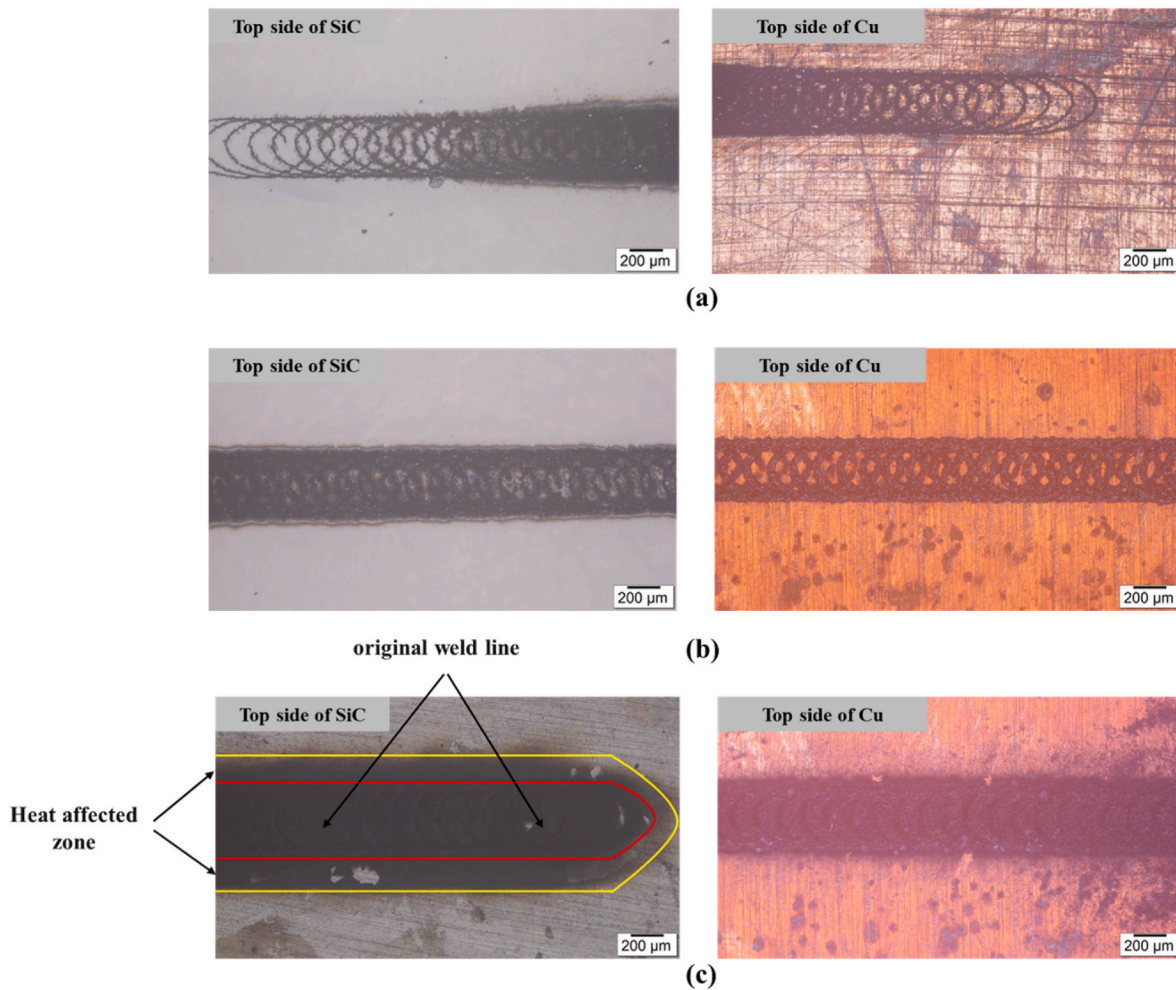


Fig. 6. Comparison between three different samples for the top side of SiC and Cu for the average laser power of (a) 28 W, (b) 38 W, and (c) 48 W, respectively, regarding a fixed scanning speed of 3 mm/s.

realistic heat distribution patterns and accurately capturing surface temperature gradients. The thermal effects depend on parameters such as the convection coefficient h_c , the ambient temperature T_0 , the emissivity factor ϵ_r , the Stefan-Boltzmann constant σ , and the evaporation velocity V_{evap} , which collectively influence the thermal behavior at these boundaries. Conversely, Eq. (8) specifies the continuity conditions for the lateral boundaries, ensuring smooth transitions and coherence in the computational domain. This setup is essential for achieving accurate thermal simulations, providing stability and consistency throughout the model.

In this research, FLOW-3D, a commercial simulation platform, was used to model the laser welding process, focusing on optimizing computational performance while ensuring precision. The model's dimensions and mesh segmentation were carefully calibrated to balance efficiency and accuracy. As shown in Fig. 5(a), variations in the domain's length and width had a minimal impact on the results. However, to align with experimental conditions, the domain thickness was precisely matched to the lap joints' thickness in the physical experiments. This adjustment ensured the simulation accurately replicated real-world scenarios, enhancing the reliability and validity of the outcomes. These targeted refinements increased the simulation's relevance, robustness, and confidence in the analysis findings. This research employed a wobbling scanning strategy as a key component of the laser welding process to enhance weld quality in dissimilar materials [36,37]. By optimizing swing width and velocity parameters, the strategy ensures uniform heat distribution, controlled penetration, and effective material

intermixing, thereby minimizing voids, brittle IMCs, and excessive HAZ formation [38–40]. This approach is particularly beneficial for brittle materials [40–42], where reducing localized thermal gradients helps maintain consistent bonding performance and prevent substrate cracking or delamination.

As shown in Fig. 5(b), a swing width "A" of 0.300 mm was established as the optimal parameter for balancing thermal distribution and penetration depth during laser welding, preserving weld integrity while reducing localized stresses that can trigger microcrack initiation in brittle SiC. In Fig. 5(b), the tangential velocity is denoted as " V_t ". A consistent overlap ratio of 70 % enhances joint strength and optimizes energy efficiency, reducing thermal stress-induced fractures in SiC. A steady travel speed, labeled " V_f " was calibrated to regulate heat input and motion stability, ensuring uniform weld formation. Thus, the dynamic movement of the heat source is mathematically described in Eq. (9), with " f " and " V_f " defined in Eqs. (10) and (11) as [5]:

$$\begin{cases} x(t) = \frac{A}{2} \cos\left(\frac{2V_t}{A}t\right) + V_f t \\ y(t) = \frac{A}{2} \sin\left(\frac{2V_t}{A}t\right) \end{cases} \quad (9)$$

and

$$f = \frac{V_t}{\pi A} \quad (10)$$

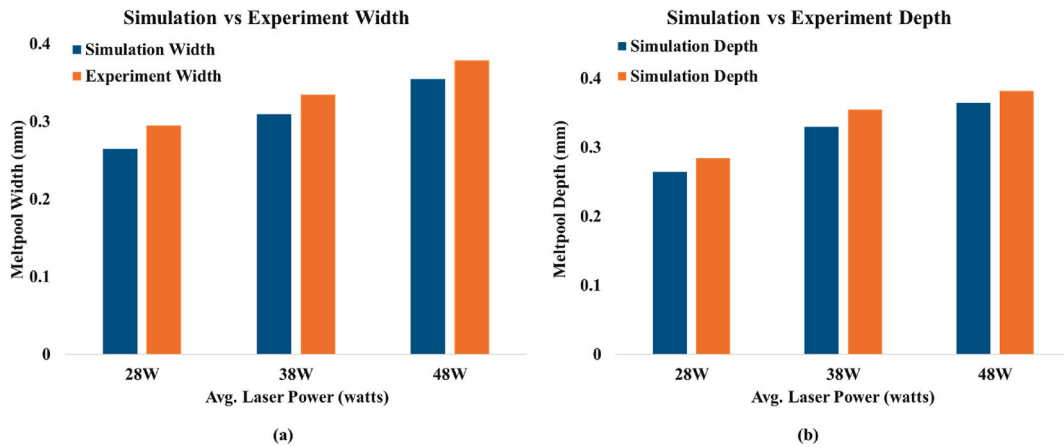


Fig. 7. Comparison of simulated and experimental melt pool dimensions at a scanning speed of 3 mm/s: (a) Melt pool width (simulation vs. experiment) and (b) Melt pool depth (simulation vs. experiment) under average laser powers of 28 W, 38 W, and 48 W.

$$V_f = f \left(A - \left(\frac{A \cdot \text{Overlap}}{100} \right) \right) \quad (11)$$

In Eqs. (9)–(11), “A” represents the oscillation amplitude — that is, half of the total oscillation width of the laser beam path in the transverse direction relative to the main scanning direction. The scanning frequency varies with the laser’s travel speed over the copper substrate. To improve precision, a point-welding approach was employed to secure the copper firmly, ensuring greater control and accuracy. This methodology enhances material stability and provides consistent weld quality.

4. Numerical model validation

Fig. 6 illustrates optical microscopy (OM) images of SiC and Cu surfaces subjected to laser processing at average laser powers of 28 W, 38 W, and 48 W, at a fixed scanning speed of 3 mm/s, highlighting differences in melting behavior, bonding quality, and HAZ characteristics. At 28 W, the laser power is inadequate to melt the SiC substrate, resulting in the absence of bonding at the SiC–Cu interface. As confirmed by the OM images, the lack of an interfacial bond indicates that this power level fails to provide sufficient thermal energy to surpass SiC’s melting threshold, thereby preventing effective adhesion with Cu. When the power is increased to 38 W, partial melting occurs on both SiC and Cu surfaces, as evidenced in the OM images. This intermediate power level delivers a balanced thermal input, facilitating the formation of a

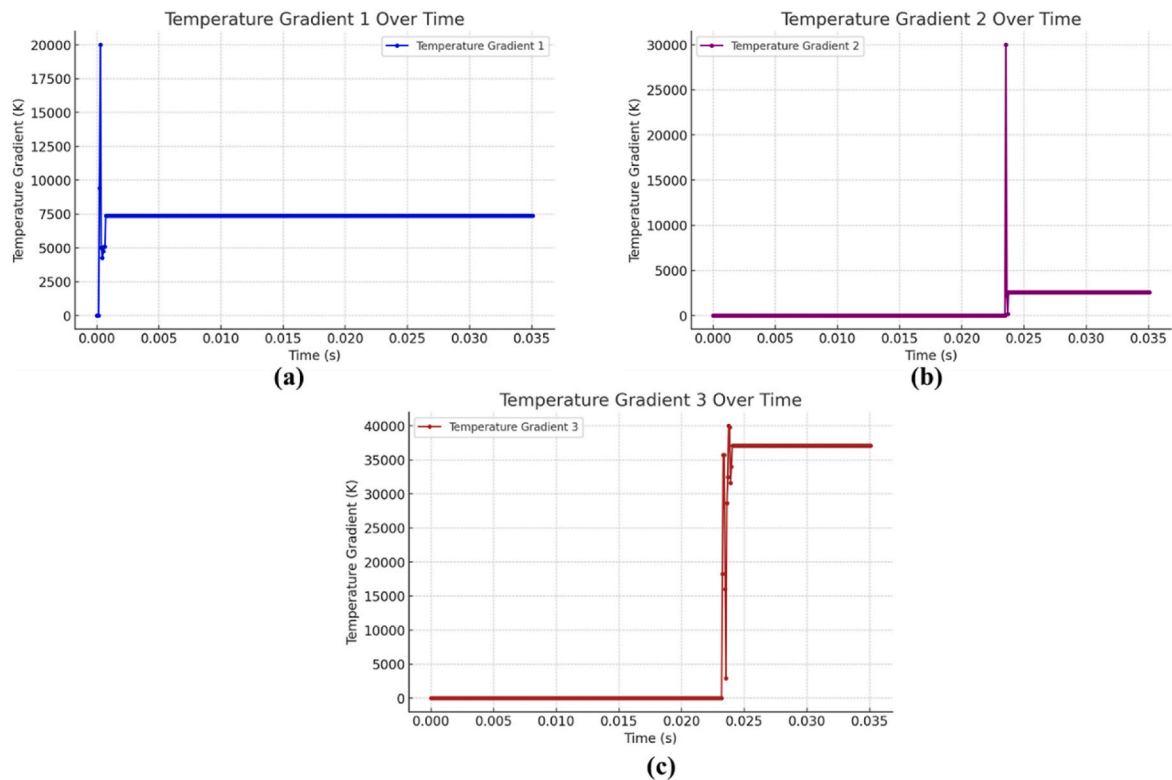


Fig. 8. Temperature gradient data from simulations at average laser powers of (a) 28 W, (b) 38 W, and (c) 48 W, respectively, regarding a fixed scanning speed of 3 mm/s. The plot shows temperature fluctuations over time, highlighting differences in temperature behavior and bonding quality across power levels.

Table 1
Laser bonding characteristics of copper and SiC at varying average laser power and scanning speed levels.

Avg. Laser Power (W)	Scanning Speed (mm/s)	Experiment Bonding Width (mm)	Experiment Bonding Depth (mm)	Experiment Ratio of Width to Depth (W/D)	Simulation Temperature Gradient (K/cm)	Region Classification
25	2	0.225	0.222	N/A	8354	No Bonding
26	2	0.244	0.245	N/A	12203	No Bonding
27	2	0.259	0.259	N/A	13001	No Bonding
28	3	0.295	0.285	N/A	20052	No Bonding
30	3	0.298	0.290	N/A	16467	No Bonding
31	3	0.310	0.295	N/A	14923	No Bonding
32	4	0.315	0.310	N/A	16862	No Bonding
33	4	0.320	0.320	N/A	17845	No Bonding
35	2	0.322	0.330	0.975	22896	Crack-Free Bonding
36	2	0.325	0.335	0.972	23082	Crack-Free Bonding
37	2	0.335	0.338	0.991	24287	Crack-Free Bonding
38	3	0.338	0.355	0.954	30078	Crack-Free Bonding
39	3	0.340	0.357	0.953	31806	Crack-Free Bonding
41	3	0.348	0.360	0.966	32817	Crack-Free Bonding
42	4	0.350	0.365	0.96	33541	Crack-Free Bonding
43	4	0.358	0.370	0.969	34505	Crack-Free Bonding
45	4	0.362	0.375	0.965	35259	Crack-Free Bonding
46	2	0.370	0.378	0.98	37380	Crack-prone
47	2	0.375	0.380	0.986	38105	Crack-prone
48	3	0.379	0.382	0.992	40000	Crack-prone
49	3	0.385	0.385	1.002	41011	Crack-Prone
50	3	0.395	0.390	1.013	42047	Crack-Prone
52	3	0.398	0.392	1.015	44402	Crack-Prone
53	4	0.405	0.395	1.025	46769	Crack-Prone
54	4	0.415	0.398	1.042	48291	Crack-Prone
55	4	0.420	0.405	1.036	49838	Crack-Prone
60	4	0.440	0.420	1.047	58344	Crack-Prone

robust interfacial bond without observable defects. The uniformity and strength of the bonding suggest that 38 W is an optimal power setting, ensuring reliable adhesion while maintaining material integrity. At 48 W, the elevated heat input produces an expanded HAZ on both the top and bottom surfaces. Cracking is observed predominantly near the fusion boundary, where temperature gradients exceed 2.8×10^4 K/cm, higher than the reported thermal shock resistance limit of SiC [43,44], inducing mechanical stress and eventual material failure. Furthermore, the significantly enlarged HAZ reflects extensive thermal diffusion into the surrounding material, which can degrade the mechanical properties of the joint.

Fig. 7 compares simulated and experimental melt pool dimensions at a fixed scanning speed of 3 mm/s, isolating the influence of laser power. Average laser powers of 28 W, 38 W, and 48 W were examined, and the resulting melt pool widths and depths are shown in Fig. 7(a) and (b). The differences between simulated and experimental values were kept within 15 %, with deviations attributed to experimental measurement uncertainties (fracture-based estimation of depth) and simplifications in the CFD model, such as temperature-dependent absorptivity and SiC decomposition kinetics. Despite these limitations, the close agreement validates the model’s predictive accuracy for melt pool geometry. Experimentally, 28 W was insufficient to achieve bonding, while 38 W produced strong interfacial bonding with minimized defect features, corroborated by shear strength results. However, defect-free bonding could not be fully confirmed due to the challenges of cross-sectional sample preparation. At 48 W, cracking and excessive HAZ expansion were observed, confirmed through fracture surface SEM and post-shear test analysis, indicating compromised structural reliability at high power. Thus, 38 W represents the most favorable condition for achieving stable bonding with reduced crack susceptibility.

4.1. Cracking criterion for laser welding of SiC–Cu

To establish a quantitative framework for predicting crack formation in SiC–Cu laser welding, we define a cracking criterion governed by the temperature gradient across the fusion boundary. The cracking criterion for SiC–Cu joints in nanosecond laser welding can be defined by the temperature gradient across the fusion boundary and the width-to-depth (W/D) ratio of the molten pool. As observed in Ref. [45] for high-strength steel, steep and uneven gradients induce thermal stresses, particularly near the fusion boundary, leading to cracking. These stresses, caused by localized laser heating and rapid cooling, result from uneven thermal expansion and contraction, often forming weak microstructures prone to cracking [46]. Materials subjected to such gradients solidify at varying rates, causing intergranular fractures or solidification cracks [47,48]. Minimizing temperature gradients through optimized welding parameters—such as reduced laser power, increased welding speed, or added heat sources—helps lower thermal stresses and mitigate cracking risks [46]. Techniques like pulsed laser welding and hybrid processes, which create uniform temperature distributions, have shown promise in reducing the effects of high gradients [49,50]. In SiC–Cu laser welding, these challenges are amplified by the differing thermal expansion properties of SiC and Cu, which increase residual stress and hinder material mixing. Uneven mixing disrupts the interface, preventing stable bond formation. This criterion moves beyond qualitative descriptions by providing threshold values that can guide parameter selection. Achieving these limits requires balancing power density and welding speed to maintain a narrow HAZ and avoid excessive thermal shock, as validated in both FLOW-3D simulations and metallographic observations.

Fig. 8 illustrates the temperature gradient plots at laser powers of 28

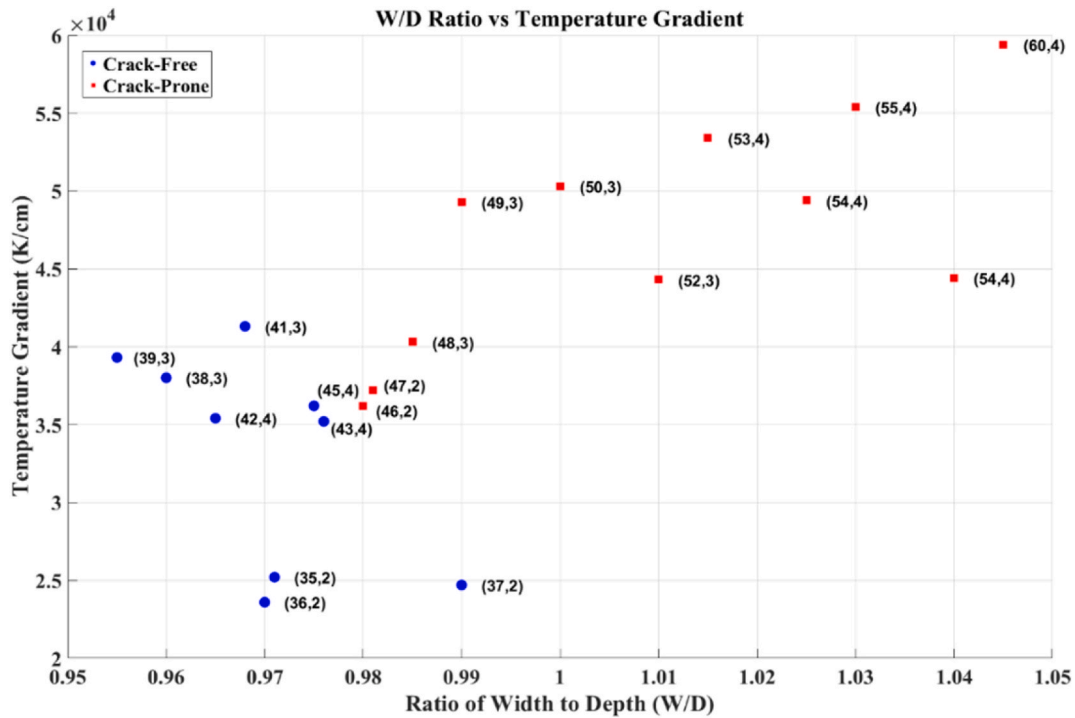


Fig. 9. Data points are labeled by process condition. The plot illustrates the trend that higher W/D and higher temperature gradient correlate with crack formation in the present study. No external threshold values are assumed in SiC–Cu Laser Welding regarding P and V .

W, 38 W, and 48 W to assess their effects on cracking and bonding in SiC–Cu joints. At a laser power of 28 W, as shown in Fig. 8(a), Temperature Gradient 1 exhibits an abrupt rise within the first millisecond, stabilizing at approximately 7500 K/cm with negligible oscillations, indicative of uniform and sustained heat flow at the probe location. At a laser power of 38 W, as shown in Fig. 8(b), Temperature Gradient 2 remains near baseline before a rapid surge to about 30000 K/cm, subsequently stabilizing at around 3000 K/cm; this transient peak is attributed to localized heat accumulation near a thermal boundary. Finally, at a laser power of 48 W, as shown in Fig. 8(c), Temperature Gradient 3 undergoes a steep increase to nearly 38000 K/cm, accompanied by minor oscillations before reaching a stable high gradient, reflecting intermittent thermal perturbations and pronounced localized absorption in the high-flux region. These findings emphasize the importance of controlling temperature gradients to prevent cracking and ensure reliable bonding. While 28 W is inadequate for bonding, 38 W achieves optimal adhesion, and 48 W introduces excessive thermal stress, causing cracks and an expanded HAZ. The distinct gradient patterns reflect variable heat distribution, affecting stress points and crack formation. Selecting appropriate laser power is critical for strong bonding without compromising material integrity at the SiC–Cu interface.

Table 1 summarizes the relationship between laser power, scanning speed, bonding dimensions, temperature gradient, and bonding classification for SiC–Cu joints, while Fig. 8 illustrates representative temperature gradient profiles. The combined data reveal three distinct regions.

- No Bonding – For laser power <35 W (temperature gradient < ~22,000 K/cm), bonding width and depth remain minimal across all scanning speeds (2–4 mm/s). Insufficient energy input prevents metallurgical bonding.
- Crack-Free Bonding – At 35–45 W with 2–4 mm/s, temperature gradients range between ~22,000 and 35,000 K/cm. This balanced thermal input achieves optimal bonding width and

depth while avoiding excessive thermal stress, producing stable, defect-free joints.

- Crack-Prone Bonding – At ≥ 46 W, temperature gradient exceeds ~37,000 K/cm (up to ~58,000 K/cm), leading to high thermal stresses that surpass the SiC–Cu interface's thermal shock resistance. Although bonding width and depth are maximized, the likelihood of crack formation increases significantly.

These thresholds, validated by both simulation and experiment, form the basis of a quantitative cracking criterion for SiC–Cu joints. The results emphasize that careful control of laser power and scanning speed is essential to maintain gradients within the 22,000–35,000 K/cm window, thereby preventing insufficient bonding at lower gradients and crack initiation at higher gradients.

4.2. Width and depth (W/D) ratio criteria for SiC–Cu joints

To complement the temperature-gradient criterion, a second cracking criterion is defined in terms of the melt pool aspect ratio (width-to-depth, W/D). The criteria for optimal melt pool depth and width in SiC–Cu laser welding were derived from established methodologies in advanced material bonding, including laser lap welding processes for glass-to-glass [51] and glass-to-Si [52] laser-welded joints. These methodologies provided a robust framework for ensuring structural integrity and minimizing the risk of cracking during welding. The benchmarks were derived through experimental validation and a simulation model to effectively predict and control melt pool characteristics. This approach establishes the W/D ratio as a quantitative indicator of crack susceptibility.

This approach was instrumental in achieving high-quality bonds with precise control over melt pool dimensions, ensuring robust and reliable joints were formed. It is noted that the bonding width (W) and depth (D) reported in Table 1 were extracted from FLOW-3D cross-sectional fields using a consistent pixel-to-length calibration and thresholding workflow. A subset of conditions was experimentally verified and overlaid as markers in Fig. 6. The OM images are used

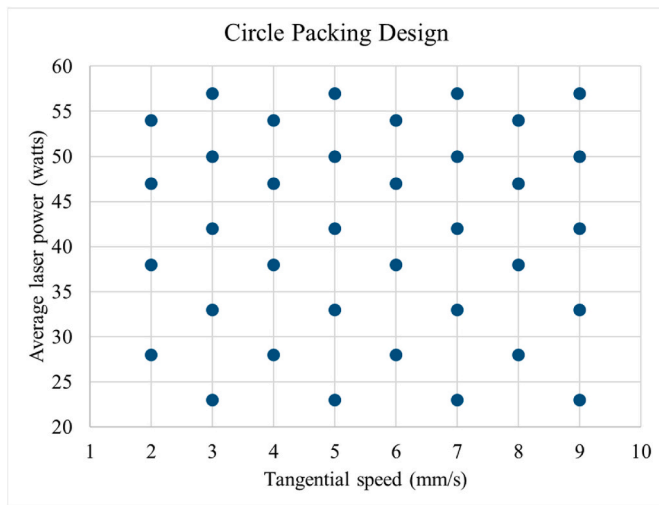


Fig. 10. Circle packing design for Simulation.

qualitatively to illustrate HAZ width and crack morphology. These measurements provided critical insights into melt pool geometry, enabling the classification of regions into "No Bonding," "Crack-Free Bonding," and "Crack-Prone" conditions. As shown in Fig. 9, cracking was observed when the W/D ratio exceeded ~ 0.967 and coincided with steep interfacial temperature gradients above $\sim 4.0 \times 10^4$ K/cm. This combination reflects shallow, wide molten pools with poor vertical heat dissipation, which amplify tensile stresses in the SiC surface and promote crack initiation. Conversely, lower W/D ratios supported deeper molten pools and improved heat transfer, reducing localized stresses and enabling crack-free bonding. The separation between "Crack-Free" and "Crack-Prone" regions shown in Fig. 9 is therefore an empirical outcome of this dataset, not a universal threshold. Accordingly, the W/D ratio was adopted as a key optimization criterion for selecting laser power and scanning speed in SiC–Cu joints.

The absence of bonding in certain regions during SiC–Cu laser welding with the wobbling scanning strategy necessitates careful consideration when analyzing W/D ratios. Although depth measurements are provided for the "No Bonding" region, these reflect

interactions only at the SiC surface, with no penetration into the Cu interface. As a result, W/D ratio calculations in these regions are irrelevant. When excluding these non-bonded cases, a threshold W/D ratio of approximately 0.967 was identified for the "Crack-Free Bonding" region, while higher ratios, averaging 1.014, correspond to the "Crack-Prone" region. This aligns with established research emphasizing that weld depth measurements should reflect actual interface fusion, which is critical for assessing weld quality [53]. Fig. 9 indicates that, within the present process window, the probability of cracking increases with both W/D and the temperature gradient. Prior studies [53,54] are cited to support the qualitative association between melt-pool aspect ratio, thermal gradients, and crack susceptibility rather than any specific numerical cutoff. Similarly, studies on keyhole and molten pool dynamics underscore the importance of analyzing bonded zones, indicating welding success [55]. By defining this threshold, the W/D ratio criterion provides a practical guideline that complements the temperature-gradient criterion (Section 4.1), together forming a dual-parameter framework for predicting crack formation in SiC–Cu laser welding.

4.3. Optimized processing map of SiC–Cu

This section describes the development of an optimized processing map for SiC–Cu laser welding, organized into two subsections. Subsection 4.3.1 details the circular packaging design model for parameter optimization. It explains the circle packing algorithm used to generate representative parameter combinations of tangential speed and average laser power. These parameters provided inputs for CFD simulations, whose results trained ANN models. These ANN models accurately predicted melt pool characteristics, including penetration width to depth ratio (W/D), and temperature gradients across the design space. Subsection 4.3.2 presents the final optimized processing map derived from ANN modeling. It emphasizes identifying optimal welding parameters that minimize defects like cracks and insufficient bonding by balancing temperature gradients, melt pool depth, and bonding width. The processing map summarizes parameter regions that, in our dataset, were most frequently associated with crack-free bonding, consistent with the temperature-gradient criterion. The map is intended as practical guidance for the present stack and optics configuration rather than a universal prescription.

4.3.1. Circular packaging design model for parameter optimization

To develop the processing map for SiC–Cu laser welding, we employed an ANN modeling framework similar to our previous studies [7,56,57]. The design space, shown in Fig. 10, was defined by two parameters: tangential speed (1–10 mm/s) of the nanosecond laser with wobbling and average laser power (20–60 W). A circle-packing algorithm in MATLAB generated 36 representative sample points, which were simulated using CFD to obtain melt pool penetration depth, bonding width, and temperature gradient. These datasets trained three ANN models, each predicting one output property. The trained models were then applied to evaluate 2997 parameter combinations across the design space. The resulting processing maps visualize the correlation between welding parameters and key outcomes—penetration depth, bonding width, and crack susceptibility. Rather than serving as a universal prescription, these maps provide a practical tool for identifying stable bonding windows and avoiding crack-prone regions in SiC–Cu welding.

4.3.2. Final optimized processing map

The ANN model used in this study was developed and trained using MATLAB's Neural Network Toolbox to establish the final optimized processing map for SiC–Cu laser welding. These models were designed to predict key melt pool characteristics, specifically penetration depth and bonding width, based on laser welding parameters such as average laser power and tangential speed. The trained ANN models systematically

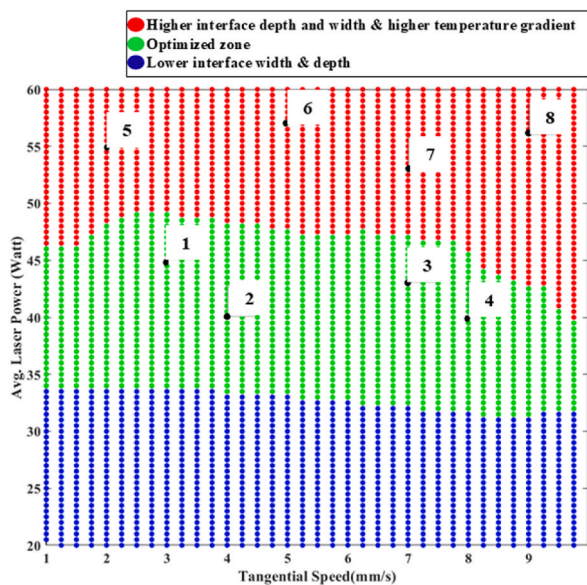


Fig. 11. Processing map for SiC–Cu laser welding showing three regimes: "No Bonding" (blue), "Crack-Free Bonding" (green), and "Crack-Prone Bonding" (red). Eight experimental points from Table 2 were used for validation.

Table 2

Selected experimental points for validating the processing map with eight representative samples.

Sample Number	Avg. Laser Power (W)	Tangential Speed (mm/s)
1	45	3
2	40	4
3	43	7
4	40	8
5	55	2
6	57	5
7	53	7
8	56	9

analyzed the processing map to identify optimal parameter ranges that ensure robust SiC–Cu joints while minimizing defects such as cracks and insufficient bonding.

As described in Section 4.1, the temperature gradient was chosen as one of the key criteria due to its critical role in influencing crack susceptibility. Excessive temperature gradients above 37,000 K/cm generate large thermal stresses at the SiC–Cu interface because of the mismatch between the brittle SiC and ductile Cu. These stresses exceed the fracture tolerance of SiC, leading to crack initiation and propagation. Conversely, insufficient gradients below 23,000 K/cm at laser powers below 35 W fail to achieve adequate melting, resulting in incomplete bonding. The optimal "Crack-Free Bonding" region occurs at intermediate temperature gradients (22,000–35,000 K/cm), achieved with laser powers of 35–45 W and scanning speeds of 2–4 mm/s. In this regime, HAZ growth was limited to ~10–25 μm , ensuring controlled thermal diffusion, stable bonding, and minimal structural damage. Melt pool dimension ratio (W/D) further validates these regions: crack-free conditions correlate with controlled geometry close to unity, whereas higher laser powers (>46 W) in the crack-prone region produce excessive melt pool depths and widths, broadening the HAZ beyond ~30 μm and increasing thermal stresses that trigger interfacial cracking.

To validate the processing map in Fig. 11, which categorizes bonding performance into three regions—"No Bonding" (blue), "Crack-Free Bonding" (green), and "Crack-Prone Bonding" (red)—as defined in

Table 1, eight representative points, detailed in Table 2, were selected for experimental verification. The "No Bonding" region (blue), characterized by insufficient thermal energy, occurs at laser powers below 35 W across tangential speeds of 1–10 mm/s, where inadequate melting prevents bonding. The "Crack-Free Bonding" region (green), the optimal range for defect-free bonding, spans laser powers of 35–45 W and tangential speeds of 1–10 mm/s. Controlled thermal gradients in this range ensure stable bonding with HAZ thickness restricted to ~25 μm . The "Crack-Prone Bonding" region (red), at laser powers above 46 W, results from excessive temperature gradients, leading to microcrack initiation in the SiC substrate and wider HAZ formation (>30 μm) despite achieving maximum melt pool size. These effects are corroborated by the OM evidence in Fig. 12.

Fig. 12 shows OM images of the top surfaces of SiC and Cu after laser processing at Sample 1 and Sample 6 conditions, illustrating how laser power influences melting behavior, bonding quality, and the HAZ. Under conditions of Sample 1, the laser induces partial melting at the SiC–Cu interface, resulting in noticeable bonding with minimal defects. The OM image in Fig. 12(a) demonstrates a uniform bonding line with HAZ width ~20 μm , confirming stable interfacial bonding with controlled heat diffusion. Under Sample 6 conditions, the increased laser power leads to a significant rise in thermal stress, exceeding the thermal shock resistance of SiC. As shown in Fig. 12(b), cracks are visible on both SiC and Cu surfaces, with HAZ expansion beyond ~30 μm , reflecting detrimental heat spread into adjacent regions. Higher power levels thus introduce excessive temperature gradients that compromise joint integrity. In summary, the findings underscore the importance of selecting optimal laser processing conditions for bonding SiC to Cu. While Sample 1 achieves effective bonding with minimal defects, Sample 6 demonstrates that excessive thermal input degrades the quality of the SiC–Cu interface. Although Fig. 12 shows only two representative cases, the validation across all eight Table 2 points confirmed the processing map predictions.

5. Results and discussion

Microstructure characterization was conducted using OM to observe

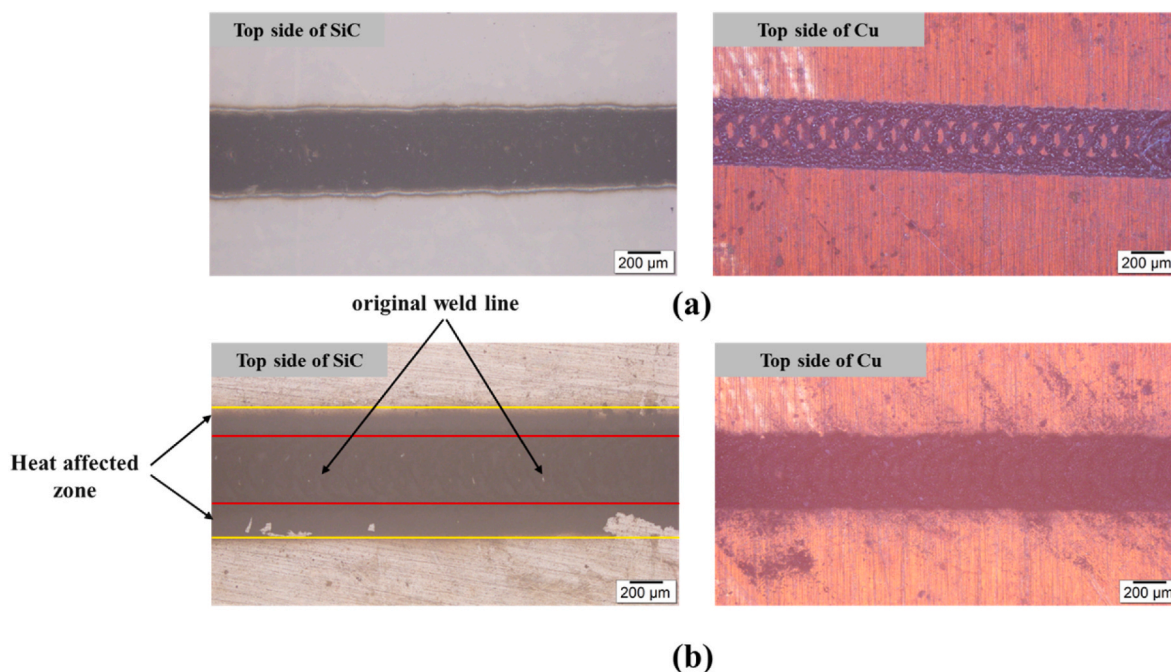


Fig. 12. OM images of the SiC (left) and Cu (right) surfaces after laser welding. (a) Sample 1 (40–45 W, 3–4 mm/s) shows defect-free bonding with a uniform interface and limited HAZ (~20 μm). (b) Sample 6 (55–57 W, 5 mm/s) exhibits visible microcracking and expanded HAZ (>30 μm), indicating excessive thermal stress.

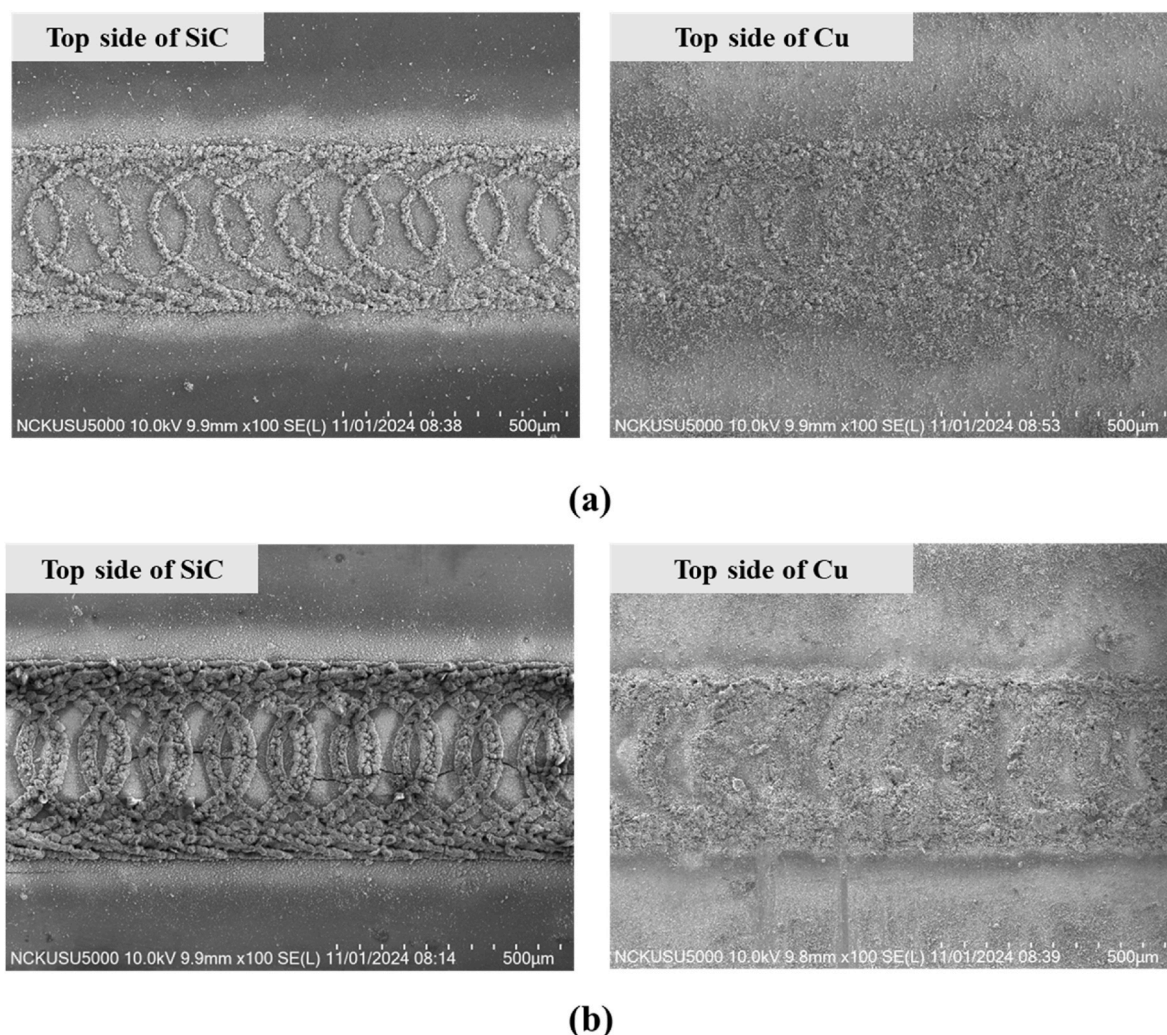


Fig. 13. Comparison between SEM images of the top surfaces of SiC and Cu after laser processing at (a) Sample 1, Avg. laser power of 45 W and scanning speed of 3 mm/s, and (b) Sample 6, Avg. laser power of 57 W and scanning speed of 5 mm/s, respectively.

weld surface morphology, SEM to assess interfacial bonding features and crack patterns, and EDS to evaluate elemental diffusion and the formation of interfacial Cu–O–Si phases (see Figs. 6 and 13–15). Fig. 13 displays SEM images of the top surfaces of SiC and Cu after laser processing under Sample 1 and Sample 6 conditions, highlighting the bonding interface and morphological changes. In Fig. 13(a), corresponding to Sample 1 (45 W, 3 mm/s), the SEM images reveal a well-defined bonding pattern with a stable interface between SiC and Cu. Partial melting at the SiC–Cu interface produced a uniform bonding line with minimal defects. The HAZ was limited to approximately 15–20 µm, indicating controlled thermal diffusion. These conditions provided sufficient thermal input for bonding while avoiding excessive stress, preserving the structural integrity of both SiC and Cu.

In Fig. 13(b), corresponding to Sample 6 (57 W, 5 mm/s), the SEM images indicate more intense laser–material interaction, leading to excessive energy input. The bonding morphology on the SiC surface shows evidence of significant surface melting and roughness, consistent with higher fusion. In contrast, the Cu surface appears smoother, reflecting its higher thermal conductivity and faster heat dissipation. However, cracks and an expanded HAZ of 30–35 µm are evident in both materials, confirming the detrimental effects of excessive thermal gradients. These features, also consistent with Fig. 12 observations, illustrate that while deeper melting was achieved, the higher stress environment compromised joint integrity and initiated interfacial cracking.

Fig. 13 emphasizes the importance of selecting appropriate processing parameters for SiC–Cu bonding. While Sample 1 achieved effective bonding with minimal defects, Sample 6 introduced excessive thermal stress, resulting in an expanded HAZ and structural degradation. These findings demonstrate that increasing laser power beyond the optimal range degrades bonding reliability by promoting crack formation and excessive heat-affected zone growth.

5.1. Results of EDS analysis

During the EDS analysis of the SiC–Cu bonded interface processed for Sample 1, distinct elemental gradients across the bonding region were observed, as shown in Fig. 14(a). The EDS analysis at selected spots captures the transition between SiC and Cu, highlighting the interfacial region's composition. EDS Spots 2, 3, 4, and 5 in the welding zone reveal copper at atomic percentages of 26.4, 28.5, 30.8, and 34.5 at. %, respectively. This distribution suggests a controlled elemental gradient that enhances joint stability by mitigating the formation of brittle IMCs [58,59]. In contrast, EDS Spots 1 and 6, located closer to the SiC substrate, revealed a higher silicon concentration of 54.5 at. % and 64.5 at. % and carbon 38.6 at. % and 28.7 at. %, with negligible presence of copper, highlighting the distinct elemental gradient between the SiC and Cu layers.

On the copper side of the interface, depicted in Fig. 14(b), EDS Spots 2, 3, 4, and 5 exhibit silicon concentrations of 28.9, 27.8, 19.8, and 32.8

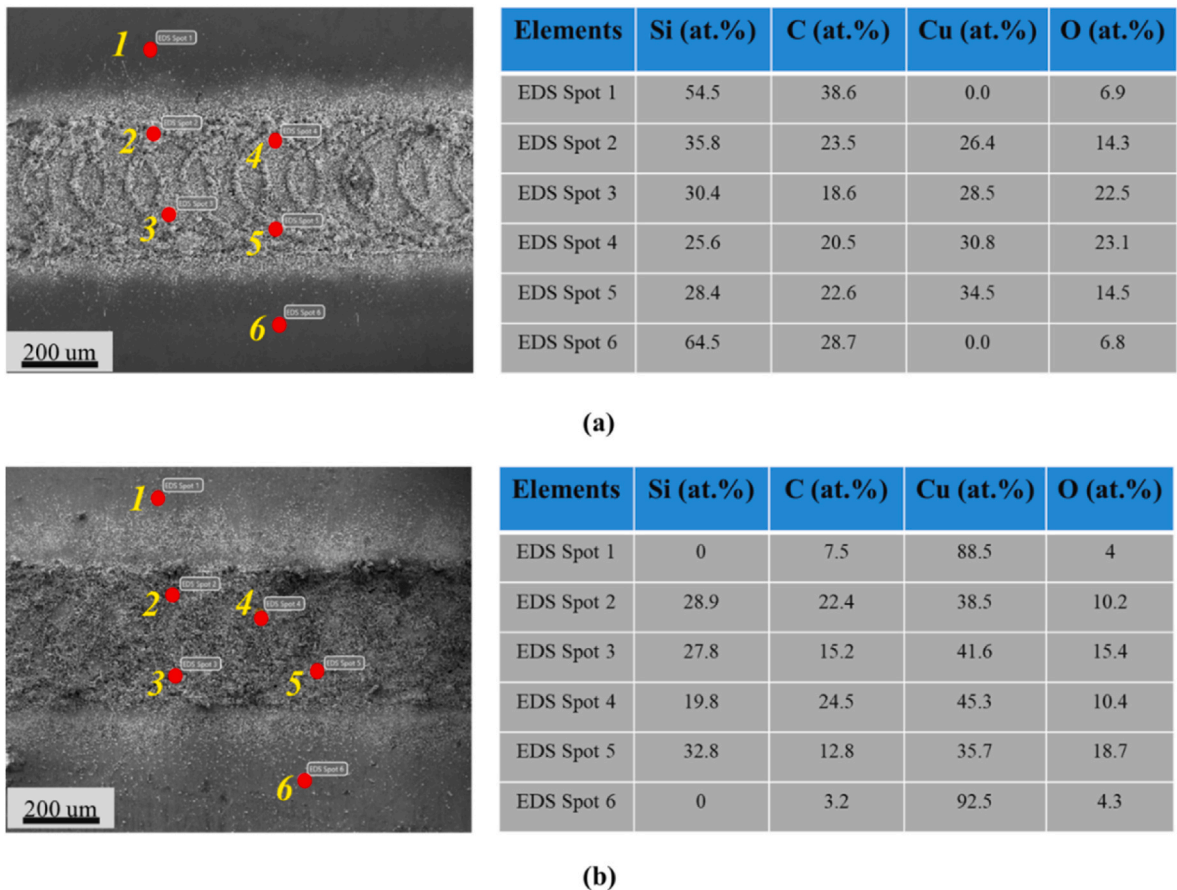


Fig. 14. EDS spot analysis of the SiC–Cu bonded interface processed upon Sample 1, showing even elemental distribution. (a) On the SiC side, Cu is evenly distributed, with higher Si and C levels near the pure SiC region. (b) On the Cu side, Si and C are present in small amounts, with Cu concentration increasing toward the pure Cu region. The quantitative spot data confirm a stable interfacial layer without signs of excessive segregation.

at. %, with corresponding carbon levels of 22.4, 15.2, 24.5, and 12.8 at. %, confirming reciprocal copper diffusion into the interfacial zone. Additionally, EDS Spots 1 and 6 on the pure copper side show high copper concentrations of 88.5 and 92.5 at. % with minimal silicon and carbon, signifying the boundary region and the presence of the pure Cu phase.

Previous studies on laser-induced plasma-assisted penetration (LIPAP) have highlighted the role of this Cu–O–Si layer in establishing strong bonds between copper and silicon-based materials [60,61]. The high localized temperatures from laser pulses facilitate copper diffusion into SiC, forming a layer enriched with crystalline copper, copper oxides, and an amorphous Cu–O–Si phase [60,61]. Similar mechanisms are inferred in this study, where the observed concentrations of copper, silicon, and oxygen suggest the presence of a bonding layer that reinforces the structural integrity of the SiC–Cu bond. The anchoring structure is further supported by the formation of nanostructures, enhancing adhesion and contributing to a structurally reinforced SiC–Cu bond.

These EDS findings underscore the robustness of the bonded interface achieved through laser processing, aligning with prior observations that SiC can decompose under high-energy laser exposure, releasing silicon and carbon species [62,63]. This decomposition likely allowed silicon to form covalent bonds with copper, as seen in Ref. [19]. The observed elemental distribution and interfacial bonding mechanisms validate the effectiveness of laser welding for durable SiC–Cu interfaces, offering promising applications in advanced engineering and manufacturing fields.

When non-optimal parameters were applied (Sample 6: 57 W, 5 mm/s), excessive laser energy resulted in stronger intermixing between SiC

and Cu, leading to irregular elemental gradients and crack formation in the SiC substrate. As shown in Fig. 15(a), EDS Spots 2, 3, 4, and 5 analyses reveal copper traces on the SiC surface, with copper atomic percentages recorded at 35.7, 70.5, 56.6, and 27.8 at. %. This higher presence of copper indicates significant mixing, contributing to structural irregularities and potential crack formation due to thermal stresses induced by high-energy laser application. In contrast, EDS Spots 1 and 6, located closer to the SiC substrate, revealed a higher silicon concentration of 59.5 and 64.6 at. % and carbon of 33.8 and 31.3 at. %, with negligible presence of copper, highlighting the distinct elemental gradient between the SiC and Cu layers.

In Fig. 15(b), the EDS analysis of the copper surface EDS Spots 2, 3, 4, and 5 shows traces of silicon and carbon, with silicon percentages of 68.3, 33.6, 47.5, and 70.6 at.% and carbon percentages of 14.4 at.%, 22.7 at.%, 14.3 at.%, and 12.9 at.% across the respective spots. This non-uniform distribution of Si and C on the copper surface leads to brittle IMCs at the bonding interface, exacerbated by the inherent brittleness of SiC and the localized thermal stresses from the laser [58,59]. Cracks observed in these regions (widths up to ~5–10 μm in localized zones) compromise the structural integrity of both materials. Additionally, EDS Spots 1 and 6 on the pure copper side showed high copper concentrations of 93.9 and 84.3 at.%, respectively, with minimal silicon and carbon, which confirms the boundary region and pure Cu phase. As shown in Fig. 14(b), the cracks compromise the material’s integrity and performance, emphasizing the need for carefully controlled parameters to ensure strong and reliable welds.

The results in Figs. 14 and 15 underscore the importance of precise parameter control in SiC–Cu laser welding. Under optimized conditions (Sample 1, Fig. 14), the bonding interface exhibited a balanced

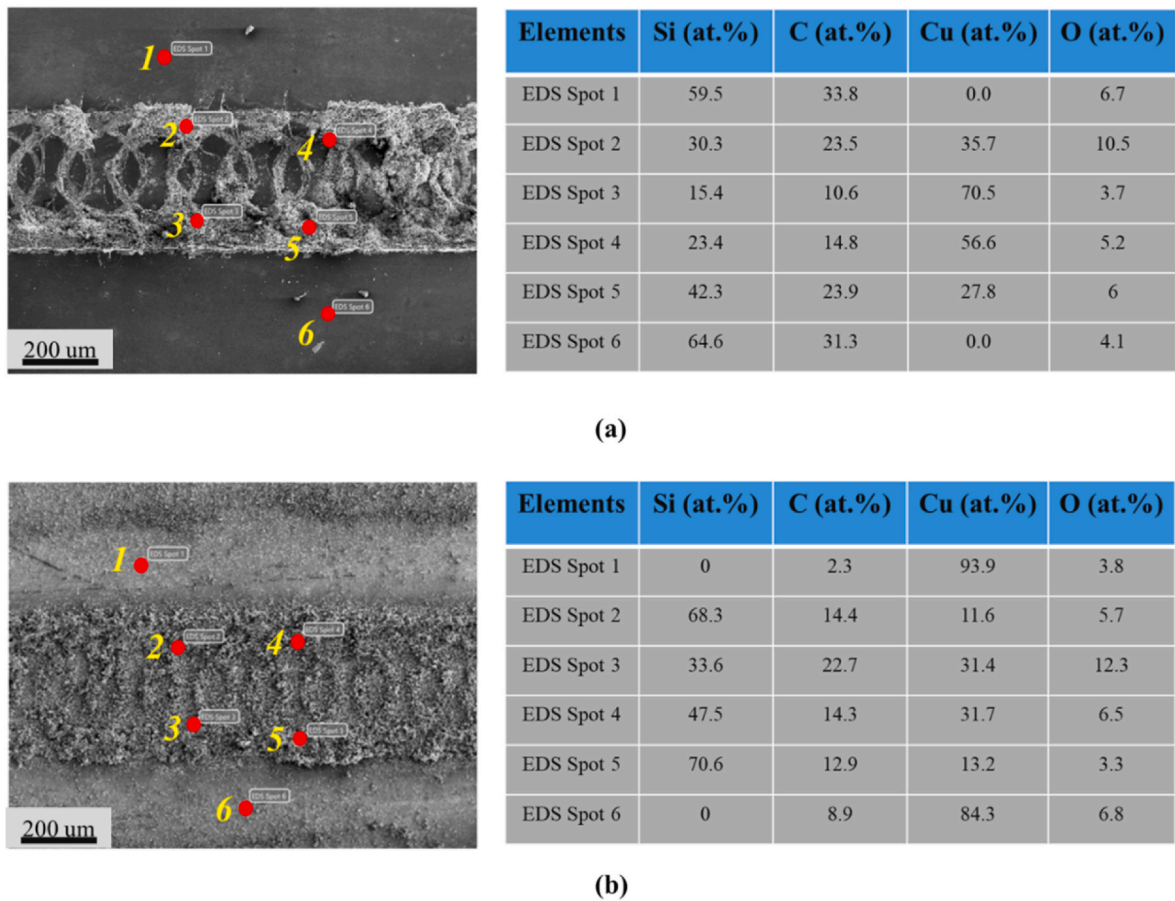


Fig. 15. EDS spot analysis of the SiC–Cu bonded interface processed based upon sample 6, showing excessive mixing and crack formation. (a) On the Cu side of the interface, Si and C appear irregularly distributed, and (b) On the SiC side, Cu is detected in localized regions, indicating excessive mixing and poor bonding quality. These results highlight the contrast between stable bonding (Sample 1) and defect-prone bonding (Sample 6).

elemental distribution, with copper concentrations on the SiC surface limited to 26.4–34.5 at.% and relatively uniform distributions of silicon (19.8–32.8 at.%) and carbon (12.8–24.5 at.%) on the Cu side. This controlled intermixing minimizes brittle IMCs [57,58] and supports stable bonding. In contrast, non-optimized conditions (Sample 6, Fig. 15) produced highly irregular distributions, with copper on the SiC surface ranging from 27.8 to 70.5 at.% and silicon on the Cu side fluctuating between 33.6 and 70.6 at.%. These variations promote brittle IMCs and crack formation, with localized cracks observed at widths up to ~5–10 μm, further exacerbated by the inherent brittleness of SiC.

Together, these findings demonstrate that moderate laser power (~45 W, 3 mm/s) maintains a stable elemental gradient and reliable bonding, whereas excessive power (~57 W, 5 mm/s) leads to excessive intermixing, interfacial instability, and fracture susceptibility. While these results clarify the role of laser parameters in joint integrity, long-term reliability under service conditions (thermal cycling, fatigue, environmental exposure) remains to be evaluated in future work.

5.2. Raman spectroscopy results for SiC– Cu irradiated surface

Complementing the EDS findings, Raman spectroscopy was employed to provide vibrational evidence of SiC decomposition and interfacial changes. On the Cu side (Fig. 16(a)), a pronounced peak at 507 cm⁻¹ (Fig. 16(c)) corresponds to crystalline Si, confirming that elemental Si released from SiC decomposition migrated and redeposited onto the Cu surface. This is consistent with the EDS results showing Si enrichment in the interfacial zones.

On the SiC surface (Fig. 16(b)), the Raman spectrum exhibits a dominant peak near 520 cm⁻¹ (Fig. 16(d)), close to the intrinsic

crystalline Si mode. Prior studies [19] have associated such features with Si-related bonding at Cu–Si interfaces, potentially influenced by interdiffusion and local strain. While this suggests possible Si–Cu interactions, the present work does not claim a definitive identification of covalent bonding. Instead, the Raman results are interpreted qualitatively to support the EDS evidence of interfacial mixing.

The focus of this study remains on processing parameter optimization and crack mitigation. A more detailed Raman-based characterization of interfacial bonding mechanisms, will be pursued in future work.

5.3. Results of shear joining strength

Shear strength Shear strength results (Fig. 17) indicate that lower laser powers (40–45 W) combined with moderate scanning speeds—corresponding to the “green zone” in the optimized processing map—produce significantly stronger and more consistent bonds, with strengths up to 4.5 MPa. Samples 1–4 demonstrate stable interfaces and low variability, confirming that balanced energy input minimizes thermal stress, promotes uniform mixing, and enhances repeatability. In contrast, samples processed at higher laser powers (53–57 W) and faster scanning speeds (Samples 5–8) fall within the “sub-optimal zone” and exhibit lower and more variable shear strengths. The reduction is attributed to excessive thermal gradients and local overheating, which promote SiC decomposition, uneven material intermixing, and crack formation at the bonding interface. These effects are particularly pronounced in Samples 5 and 6, where cracks and irregular bonding morphology directly reduce mechanical integrity. Sample 8 exhibited the highest variability, underscoring that surpassing critical thermal thresholds compromises bond stability.

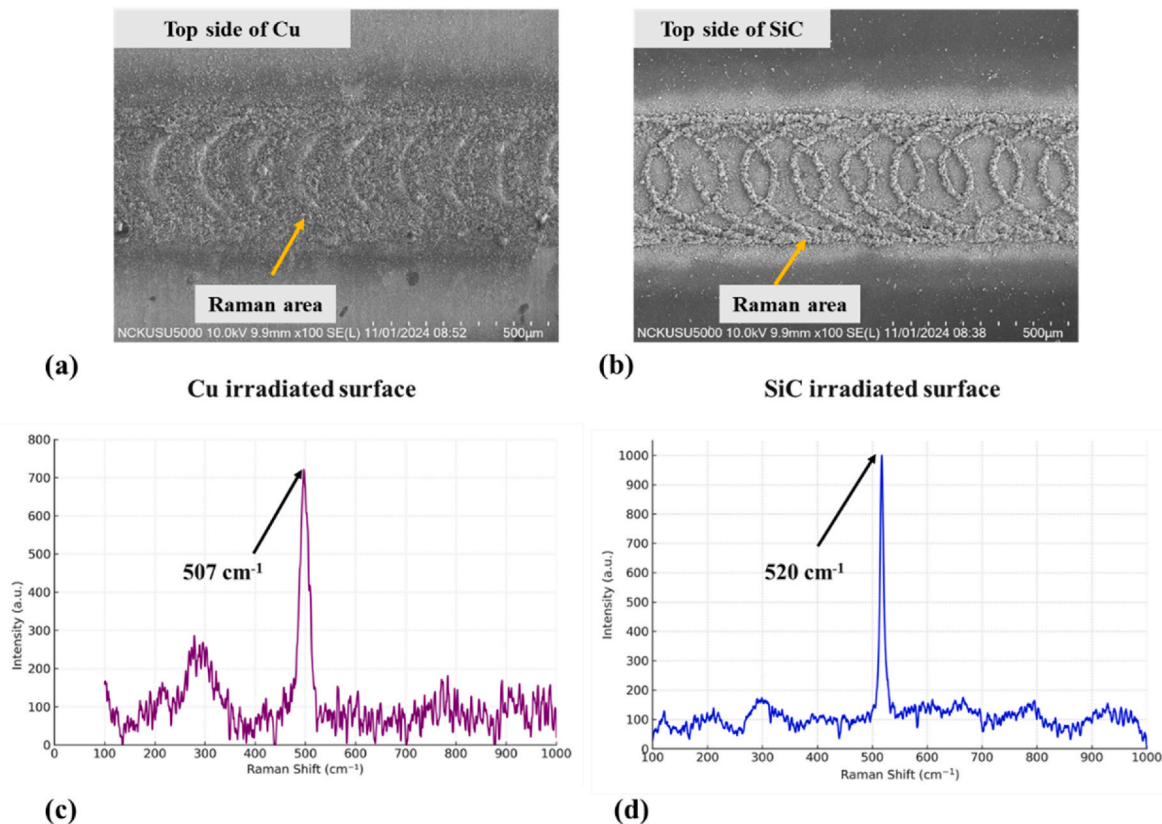


Fig. 16. SEM images of the laser-irradiated regions on (a) the Cu side and (b) the SiC side, with yellow arrows indicating the Raman analysis zones. Corresponding Raman spectra for (c) the Cu-irradiated surface, and (d) the SiC-irradiated surface reveal characteristic peaks at 507 cm^{-1} and 520 cm^{-1} , respectively.

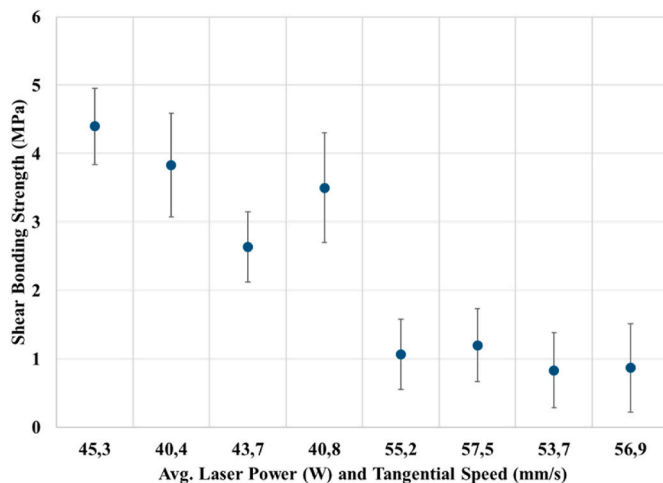


Fig. 17. Shear bonding strength of SiC–Cu joints under eight processing conditions. Samples 1–4 (“green zone”) show higher and more consistent strength, while Samples 5–8 (“sub-optimal zone”) exhibit reduced values and greater variability due to excessive thermal gradients and cracking.

Overall, the results demonstrate that reliable bonding is achieved within the “Crack-Free” region of the processing map (35–45 W, 2–4 mm/s), where controlled energy input suppresses defects and limits SiC decomposition. These findings align with the cracking criteria presented in Sections 4.1 and 4.2, confirming that temperature gradient control and W/D ratio optimization are essential for achieving consistent shear bonding strength.

5.4. Comparisons of our previous study and the other groups

Compared to our previous work on Cu–SiC bonding with Cu placed on top [30], the present study investigates the reversed configuration with SiC positioned on top, which exposes the brittle ceramic directly to the incident laser irradiation. This configuration amplifies the influence of temperature gradients on crack formation, leading to lower shear strength ($\sim 4.5\text{ MPa}$) relative to the Cu-on-top case. To address this, the present work introduces two quantifiable criteria absent from earlier studies: a temperature gradient threshold of $\sim 22,000\text{--}35,000\text{ K/cm}$ and a width-to-depth (W/D) ratio of ~ 0.967 , both of which distinguish crack-free from crack-prone regimes. Additionally, an ANN-guided processing map was developed to optimize parameter windows for stable bonding, advancing beyond prior studies on Al–Cu [7,55], Al-based SLM [56], and Cu–SiC joints [30].

During nanosecond laser irradiation, localized heating induces SiC decomposition, releasing Si and C species that diffuse into copper and facilitate the formation of an interfacial Cu–O–Si layer, composed of crystalline copper, copper oxides, and amorphous Cu–O–Si phases [59, 60]. While this interfacial layer enhances structural cohesion, excessive decomposition at temperatures exceeding $\sim 2800\text{ K}$ [64,65], reduces density and weakens interfacial integrity by releasing volatile species. Studies on laser-induced phase separation [63,66] further confirm that steep thermal gradients exacerbate decomposition and promote uneven mixing, thereby increasing crack susceptibility. Taken together, these findings emphasize the critical role of joint configuration and thermal management. By maintaining processing conditions within the defined “green zone,” the present study provides a systematic framework to mitigate SiC decomposition and crack formation, enabling strong and reliable SiC–Cu joints for advanced applications.

Table 3 summarizes laser joining studies of material combinations, highlighting key findings and challenges. Our current study focuses on

Table 3
Summary of laser joining studies on material combinations, highlighting key findings and challenges addressed.

Research Group	Material Combination	Pulse Duration	Wavelength (nm)	Shear Strength (MPa)	Key Findings	Challenges Addressed
Chambonneau et al. [19], 2021	Silicon–Copper	9.8 ps	1555	2.20	Pre-compensation of the nonlinear focal shift improves shear strength significantly.	Controlling the nonlinear propagation effects and optimizing the focal position.
Nguyen et al. [61], 2024	Quartz Glass–Copper	100 ns	1064	33.40	Metallurgical bonding and mechanical interlocking contribute to high shear strength.	Large coefficient of thermal expansion differences and microcrack prevention.
Zhang et al. [17], 2015	Alumina–Silicate Glass–Copper/ Aluminum/Steel	160 fs	800	2.34	Micron/nanometer-sized metal particles produced by laser ablation act as an adhesive.	No intermediate layer or pressure assistance needed.
Our previous study [30], 2025	Copper–Silicon Carbide	120 ns	1064	20.00	Achieve crack-free Cu–SiC joints via ANN-guided nanosecond laser wobbling, validated by cohesive fracture and uniform Cu–O–Si interfacial bonding.	Overcame brittle fracture and interfacial mixing issues in Cu–SiC welding by optimizing melt pool geometry and suppressing IMC formation.
Current Study, 2025	Silicon Carbide–Copper	120 ns	1064	4.50	Temperature gradient control (22,000–35,000 K/cm), Cu–O–Si interface formation.	Managing temperature gradients and preventing cracking.

SiC–Cu bonding using a nanosecond laser, addressing temperature gradient control and interfacial adhesion improvement. Our current study investigates the laser joining of SiC and Cu using a nanosecond laser (120 ns, 1064 nm), achieving a shear strength of 4.5 MPa. This represents a significant improvement compared to prior ultrafast laser [19] studies on silicon-copper bonding (2.2 MPa). A key contribution of the current work is the precise control of temperature gradients (22,000–35,000 K/cm), which correlates with improved interfacial compositional uniformity. Phase-specific identification of any oxygen-containing interfacial products was not performed here and will be pursued via TEM and higher temperature reliability in future work. Unlike earlier investigations on quartz glass–copper [61] and alumina-silicate glass-metal [17] joints, which relied on experimental techniques such as metallurgical bonding or laser-induced nanostructures, our work employs a simulation-based approach to optimize heat distribution, thereby mitigating thermal expansion mismatches and cracks while reducing the need for costly experimental iterations. These findings provide a novel pathway for improving laser-assisted metal-ceramic joining.

6. Conclusions

This study examined the influence of nanosecond laser welding parameters on the bonding and cracking behavior of SiC–Cu joints through combined experimental and computational approaches. The results identify optimal temperature gradients of 22,000–35,000 K/cm for crack-free bonding, while gradients exceeding 37,000 K/cm promote thermal stress and crack initiation. These findings establish quantitative criteria for process optimization to achieve a defect-free SiC–Cu joint. The optimal W/D ratio of approximately 0.967 ensures defect-free bonding, whereas higher ratios (~1.014) are associated with crack formation. A “crack-free bonding” window was identified at laser powers of 35–45 W and scanning speeds of 1–10 mm/s, where balanced thermal input effectively limits HAZ expansion. Exceeding 46 W elevates temperature gradients beyond 37,000 K/cm, leading to non-uniform material mixing, enlarged HAZ, crack initiation, and a significant reduction in shear strength to ~1 MPa. These results underscore the necessity for precise laser parameter control to mitigate thermal stress and preserve joint integrity. Shear strength testing confirmed that the highest joint strength of 4.5 MPa was achieved at 45 W and 3 mm/s, reinforcing the critical role of controlled thermal input in maximizing mechanical performance. The EDS analysis confirmed that under the optimized laser parameters, a continuous Cu–O–Si interfacial layer was formed, which was associated with reduced presence of brittle intermetallic phases and improved joint

integrity. This work demonstrates that controlling temperature gradients within the experimentally validated range and applying appropriate thermal input can achieve crack-free, mechanically robust SiC–Cu nanosecond laser welds. The results provide experimentally supported guidelines for parameter selection to enhance joint reliability in semiconductor–metal assemblies, with potential relevance to advanced applications.

AI disclosure statement

During the preparation of this work, ChatGPT was utilized to refine the writing style, improve clarity, and ensure proper grammar and spelling. The AI tool was not used to generate or analyze any data presented in this work. The authors subsequently reviewed and edited the content to align with the intended academic rigor and take full responsibility for the final content of this publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thankfully acknowledge the support provided to this study by the Ministry of Science and Technology of Taiwan under Grant No. MOST 111–2218-E-006–016 and the National Science and Technology Council of Taiwan under Grant No. NSTC 114-2640-E-006-006.

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